

# Testing the same covariates at the cell level inflates significance without changing effect size

One grey segment per covariate and dimension, joining the two ways of testing it. Dashed line, adjusted  $p = 0.05$ .

Segments are near-vertical: repeating a sample's value across its cells leaves the variance explained almost unchanged

while moving the  $p$ -value by tens of orders of magnitude, because the added rows are copies rather than independent observations.

● grouped (donor / sample) ● cell level (sample value repeated per cell)

